Multi-Branch Enterprise Network Design

A Cisco Packet Tracer WAN Simulation Project Report

Done by

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1. Introduction

This report documents the design and implementation of a multi-branch enterprise network, simulated in Cisco Packet Tracer. The company operates a central Headquarters (HQ) in Dhaka and two branch offices located in Chattogram and Sylhet. Although geographically separated, all three sites are interconnected across a Wide Area Network (WAN) so that they function as a single, unified corporate network.

Each site maintains its own local network (LAN) with departmental VLANs and local DHCP, while a dynamic routing protocol (OSPF) ensures that every site automatically learns how to reach every other site. This design is scalable: additional branches can be added later with minimal reconfiguration.

2. Project Objectives

- Connect a Headquarters and two branch offices into one enterprise-wide network.
- Assign non-overlapping, hierarchical IP addressing across all sites.
- Establish WAN links between HQ and each branch using point-to-point connections.
- Configure OSPF dynamic routing so all sites share reachability information automatically.
- Verify end-to-end connectivity between users at different physical sites.

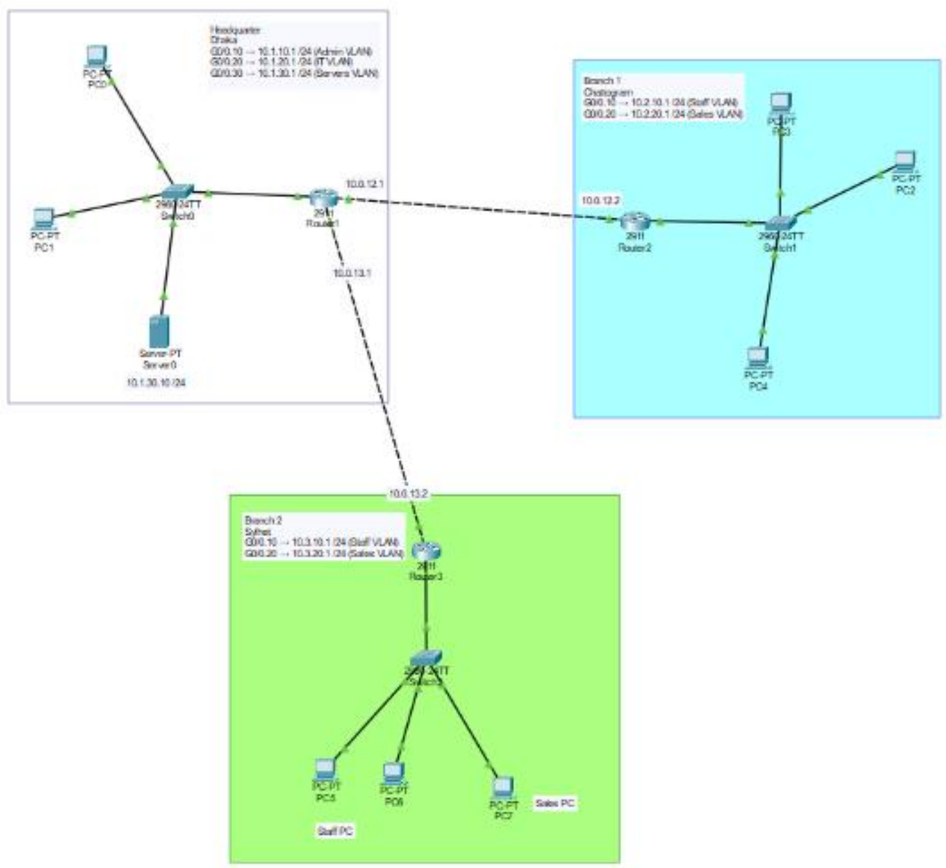


Figure 1Multi-Branch Enterprise Network Design

3. WAN Topology Overview

The network uses a hub-and-spoke topology. HQ acts as the central hub, and each branch office connects to HQ over a dedicated WAN link (the spokes). All inter-branch traffic is routed through HQ. This is the most common enterprise WAN model because it is simple to manage, cost-effective, and easy to scale.

Logical layout:

- HQ (Dhaka): central hub. Hosts corporate servers and IT administration.
- Branch 1 (Chattogram): spoke, connected to HQ via WAN link 1.
- Branch 2 (Sylhet): spoke, connected to HQ via WAN link 2.

Equipment per site:

- 1 × Router (Cisco 2911) : LAN gateway, WAN link, and OSPF routing.
- 1 or more × Switch (Cisco 2960) : connects local end devices and VLANs.
- PCs and (at HQ) a server for corporate services.

4. Site & LAN Addressing Plan

The 10.0.0.0/8 private range is divided so each site receives its own /16 block. This keeps addressing organized, prevents overlap, and makes routing tables clean and summarizable.

Site	Role	LAN Block	VLAN / Subnet	Gateway
HQ (Dhaka)	Headquarters	10.1.0.0/16	Admin — 10.1.10.0/24	10.1.10.1
HQ (Dhaka)	Headquarters	10.1.0.0/16	IT — 10.1.20.0/24	10.1.20.1
HQ (Dhaka)	Headquarters	10.1.0.0/16	Servers — 10.1.30.0/24	10.1.30.1
Branch 1 (Ctg)	Branch office	10.2.0.0/16	Staff — 10.2.10.0/24	10.2.10.1
Branch 1 (Ctg)	Branch office	10.2.0.0/16	Sales — 10.2.20.0/24	10.2.20.1
Branch 2 (Syl)	Branch office	10.3.0.0/16	Staff — 10.3.10.0/24	10.3.10.1
Branch 2 (Syl)	Branch office	10.3.0.0/16	Sales — 10.3.20.0/24	10.3.20.1

5. WAN Link Addressing

Each WAN link between two routers is a point-to-point connection using a /30 subnet, which provides exactly two usable host addresses — one for each router. A separate 10.0.x.x range is reserved for these links to keep them distinct from LAN traffic.

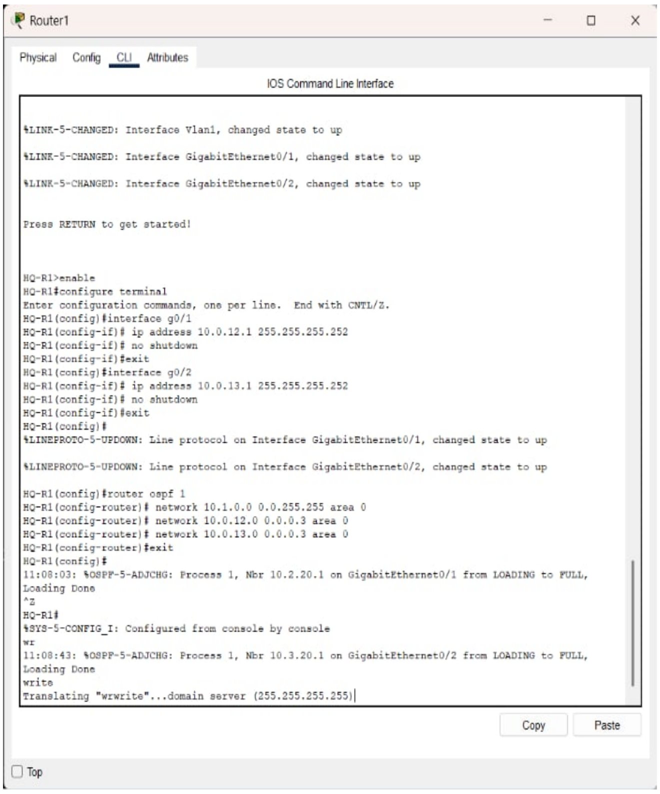
WAN Link	Subnet	HQ Interface	Branch Interface
HQ ↔ Branch 1	10.0.12.0/30	G0/1 — 10.0.12.1	G0/1 — 10.0.12.2
HQ ↔ Branch 2	10.0.13.0/30	G0/2 — 10.0.13.1	G0/1 — 10.0.13.2

6. Dynamic Routing with OSPF

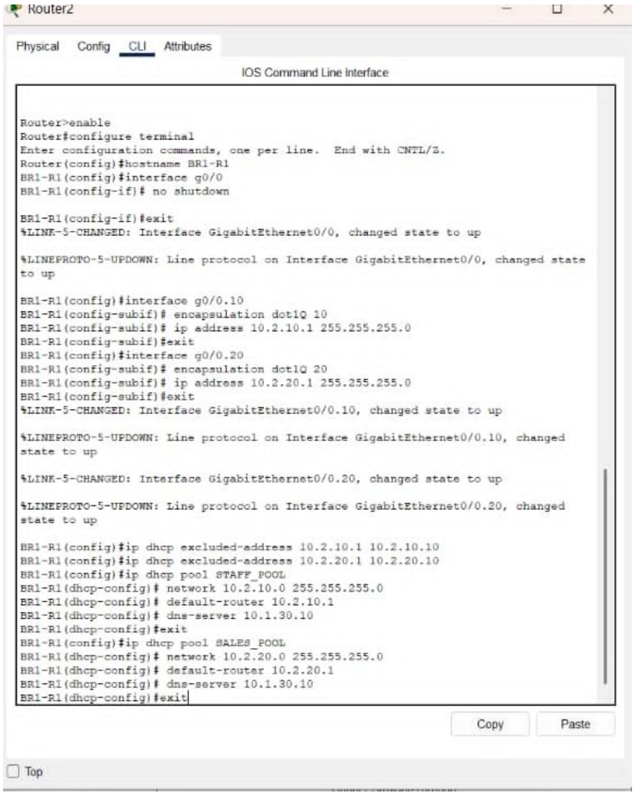
OSPF (Open Shortest Path First) is used as the routing protocol. Each router advertises its local LAN networks and its WAN links into OSPF Area 0. OSPF then floods this information so every router builds a complete map of the enterprise and automatically calculates the best path to every subnet. Unlike static routing, no manual route entries are needed, and adding a new branch requires only enabling OSPF on the new router.

7. Key Configuration Commands

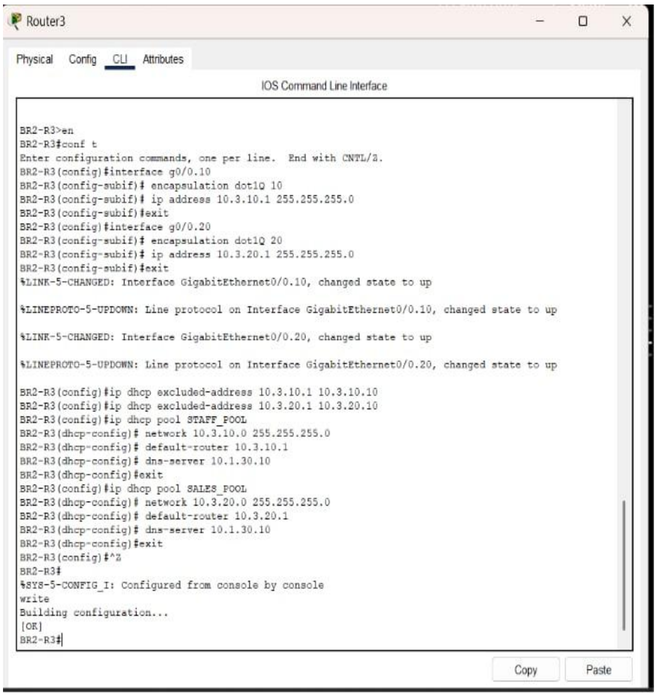
7.1 HQ Router (Hub) — HQ router configuration, sub-interface per VLAN, WAN Interfaces & OSPF

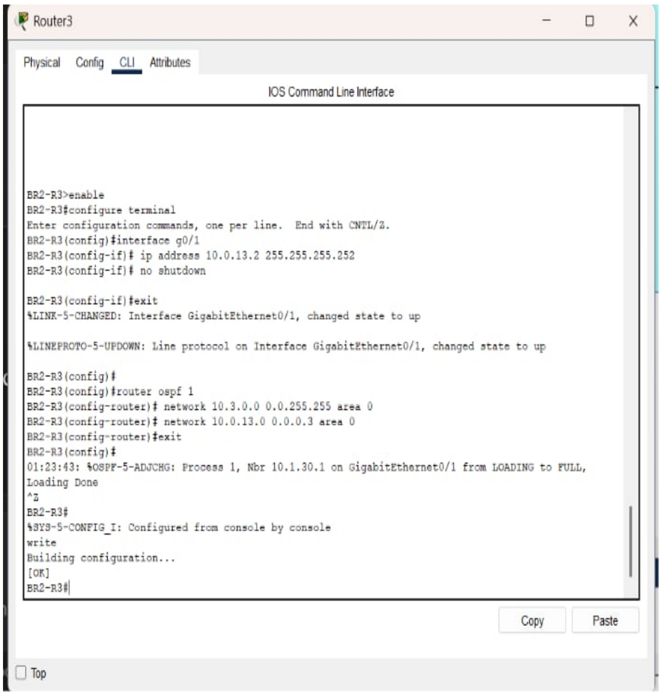


7.2 Branch 1 Router (Chattogram) — WAN & OSPF



7.3 Branch 2 Router (Sylhet) — WAN & OSPF





8. Testing & Verification

The following commands confirm that all sites form one working network:

- show ip ospf neighbor –

```
HQ-R1>
HQ-R1>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.2.20.1         1   FULL/DR         00:00:37    10.0.12.2       GigabitEthernet0/1
10.3.20.1         1   FULL/DR         00:00:37    10.0.13.2       GigabitEthernet0/2
HQ-R1>
```

- show ip route ospf –

```
HQ-R1>
HQ-R1>show ip route ospf
     10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
O       10.2.10.0 [110/2] via 10.0.12.2, 00:48:10, GigabitEthernet0/1
O       10.2.20.0 [110/2] via 10.0.12.2, 00:48:10, GigabitEthernet0/1
O       10.3.10.0 [110/2] via 10.0.13.2, 00:48:10, GigabitEthernet0/2
O       10.3.20.0 [110/2] via 10.0.13.2, 00:48:10, GigabitEthernet0/2

HQ-R1>
```

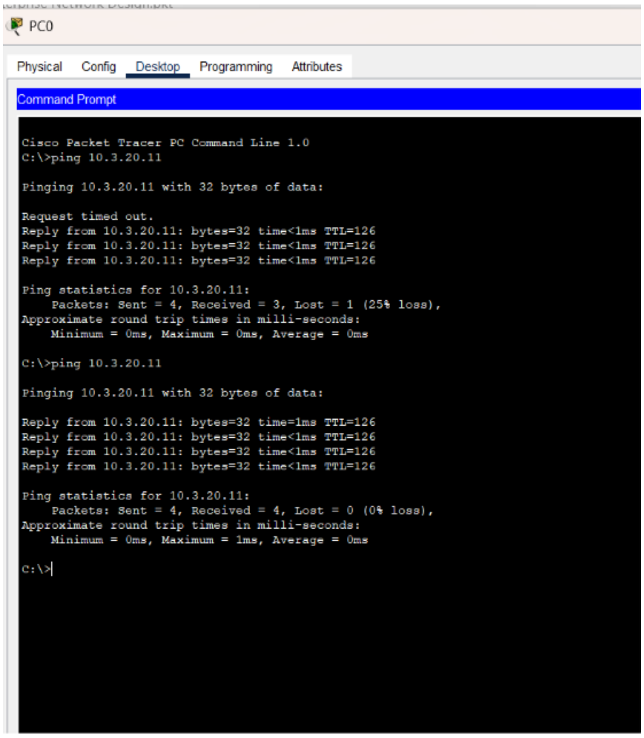
```
BR1-R1>
BR1-R1>show ip route ospf
     10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
O       10.0.13.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.10.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.20.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.30.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.3.10.0 [110/3] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.3.20.0 [110/3] via 10.0.12.1, 00:49:23, GigabitEthernet0/1

BR1-R1>
```

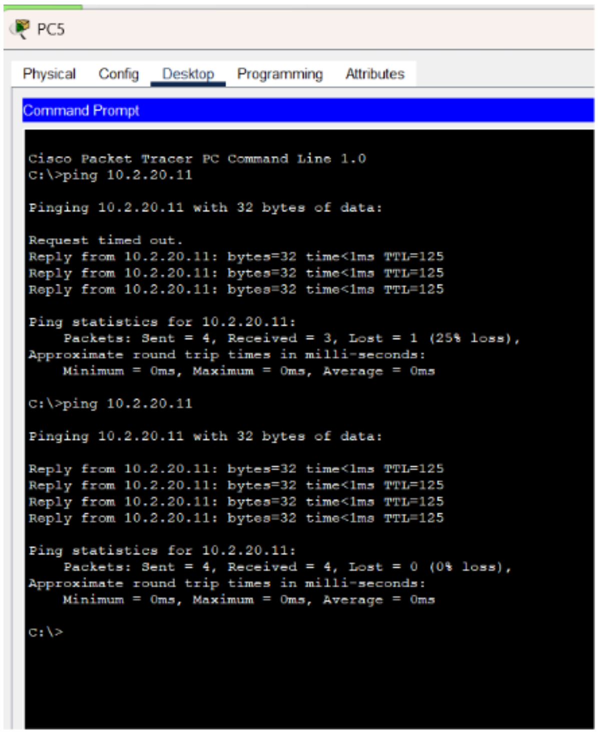
- show ip interface brief - confirms WAN interfaces are up/up.

```
BR1-R1>
BR1-R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.2.10.1       YES manual up                    up
GigabitEthernet0/0.20  10.2.20.1       YES manual up                    up
GigabitEthernet0/1     10.0.12.2       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
BR1-R1>
```

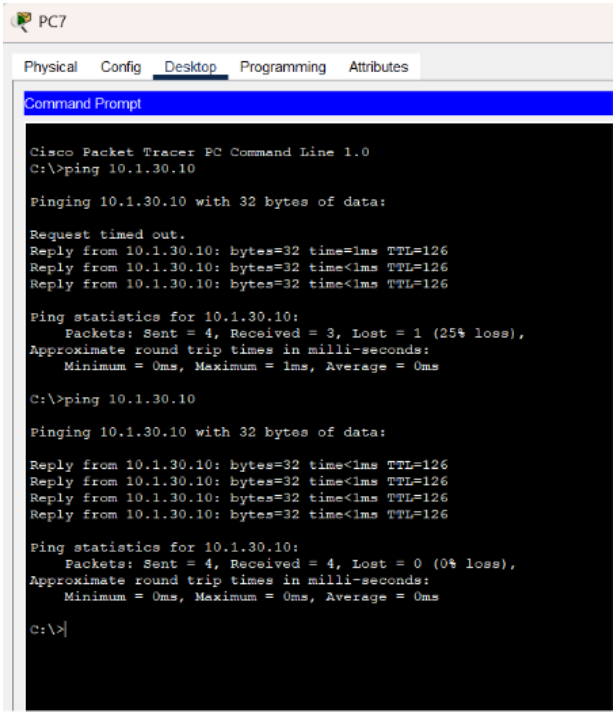
```
BR2-R3>
BR2-R3>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.3.10.1       YES manual up                    up
GigabitEthernet0/0.20  10.3.20.1       YES manual up                    up
GigabitEthernet0/1     10.0.13.2       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
BR2-R3>
```

- Cross-site ping –





- Ping to HQ server-



9. Conclusion

The project demonstrates a scalable multi-branch enterprise network in which three geographically separate sites operate as a single unified network. Hierarchical /16 addressing keeps each site organized, point-to-point WAN links interconnect the offices, and OSPF dynamic routing automatically maintains full reachability between all sites. The design scales cleanly: a new branch can be integrated simply by adding a WAN link to HQ and enabling OSPF. Future enhancements could include inter-site access-control policies, WAN link redundancy, and per-department DHCP at each branch.